

# Hardware Implementation of Convolutional Neural Network for ECG Analysis and Classification

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## MOTIVATION

- Cardiovascular Whydisease: #1 cause of death worldwide — 17.9 million deaths per year
- ECG is the primary diagnostic tool — but manual reading is slow and requires specialists
- FPGAs enable real-time, deterministic, low-power AI inference at the edge — no cloud, no CPU
- Goal: Classify arrhythmias in **2.2 ms** on a \$200 FPGA board

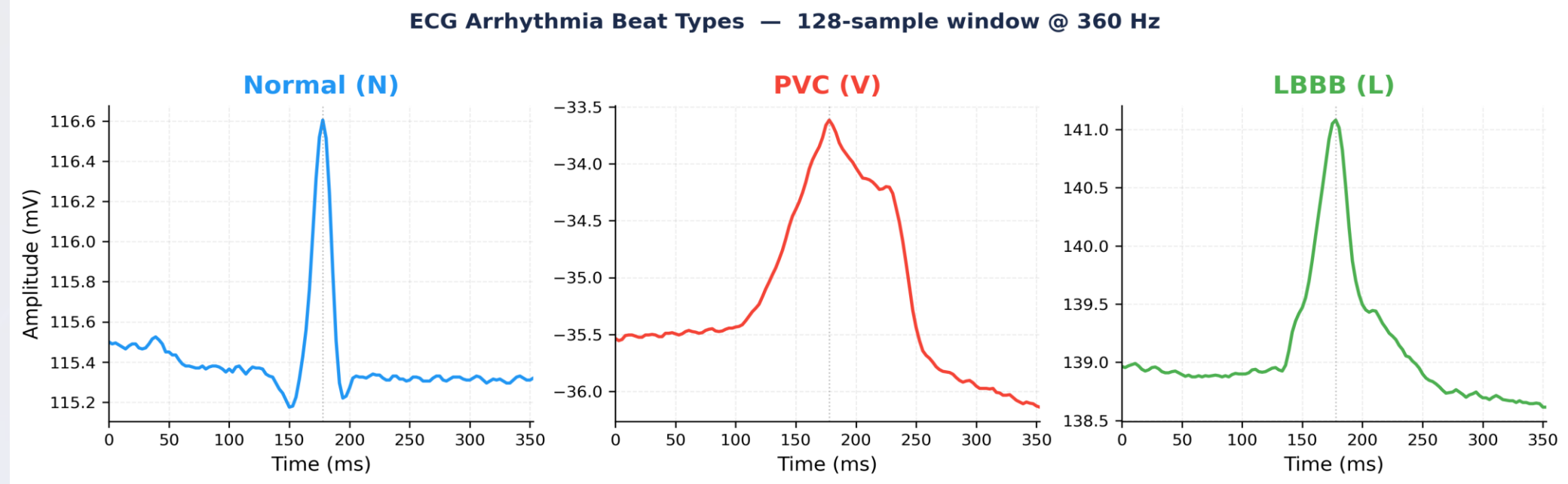


Figure 1: Representative 128-sample beat windows for three arrhythmia classes.

## APPROACH

### System Architecture

- Python streams ECG records at 360 Hz via RS-232 UART from PC to FPGA
- CNN classifies 128-sample beat windows entirely in hardware
- Result displayed on LCD ("Normal" / "Abnormal") and green LED array
- No external memory — all 18 weight files stored in on-chip M4K ROM

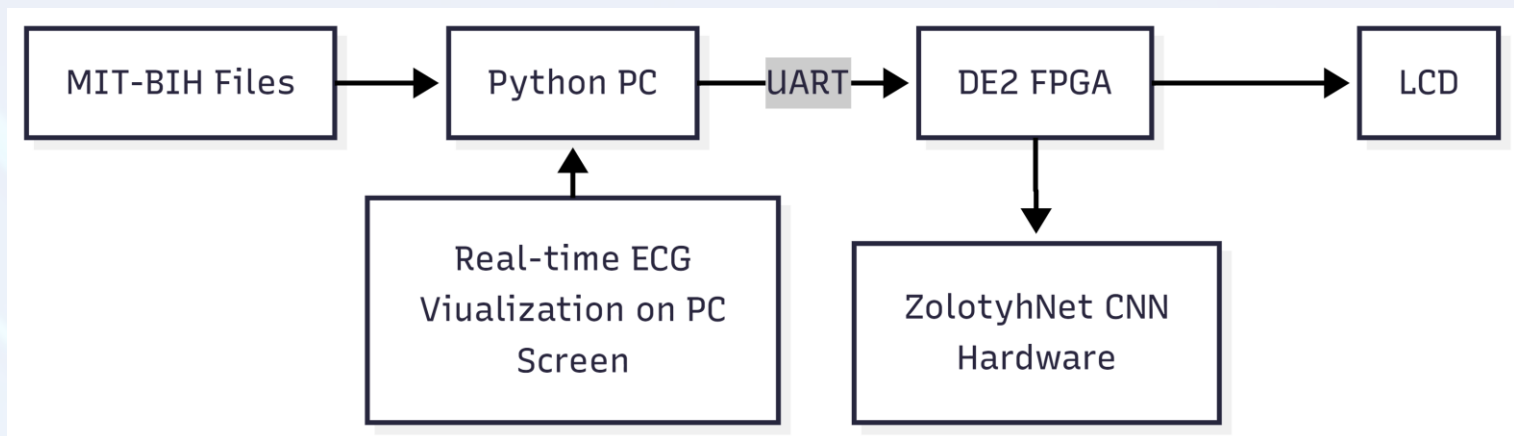


Figure 2: System block diagram.

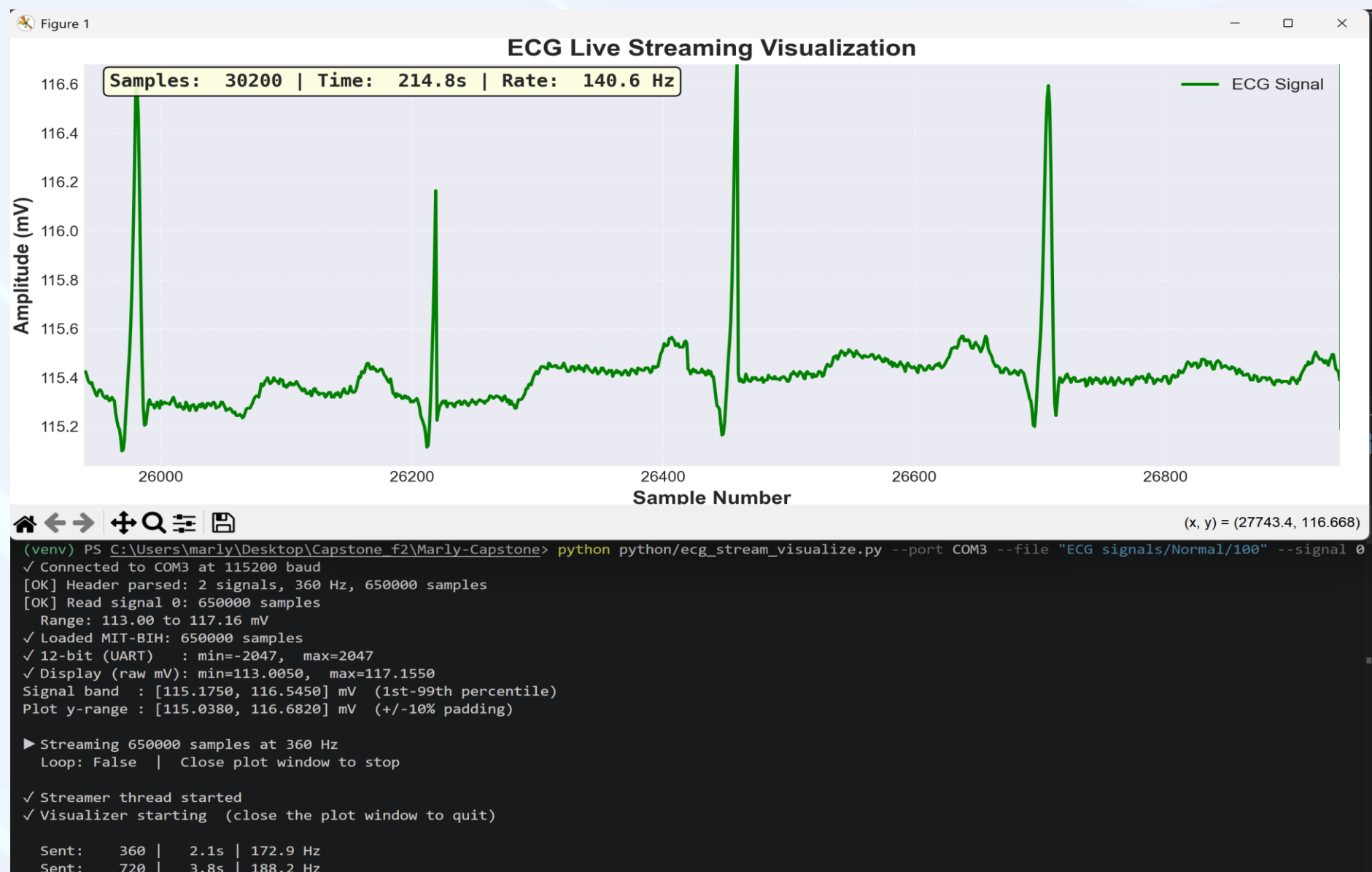


Figure 3: Python ECG visualizer displaying a scrolling real-time ECG waveform.

### CNN Architecture

- Dual-path design:** Conv1D upper path (local morphology) + FC lower path (global statistics)
- 5 Conv1D layers** + 4 MaxPool → **3 Linear layers** → element-wise ADD fusion → Argmax
- ~14,700 parameters** — 34× smaller than EcgResNet34, fits entirely in on-chip ROM
- Trained on MIT-BIH Arrhythmia Database — 9,368-sample validation set
- Weights quantized to **Q8.8 fixed-point** (16-bit) — no floating-point hardware needed

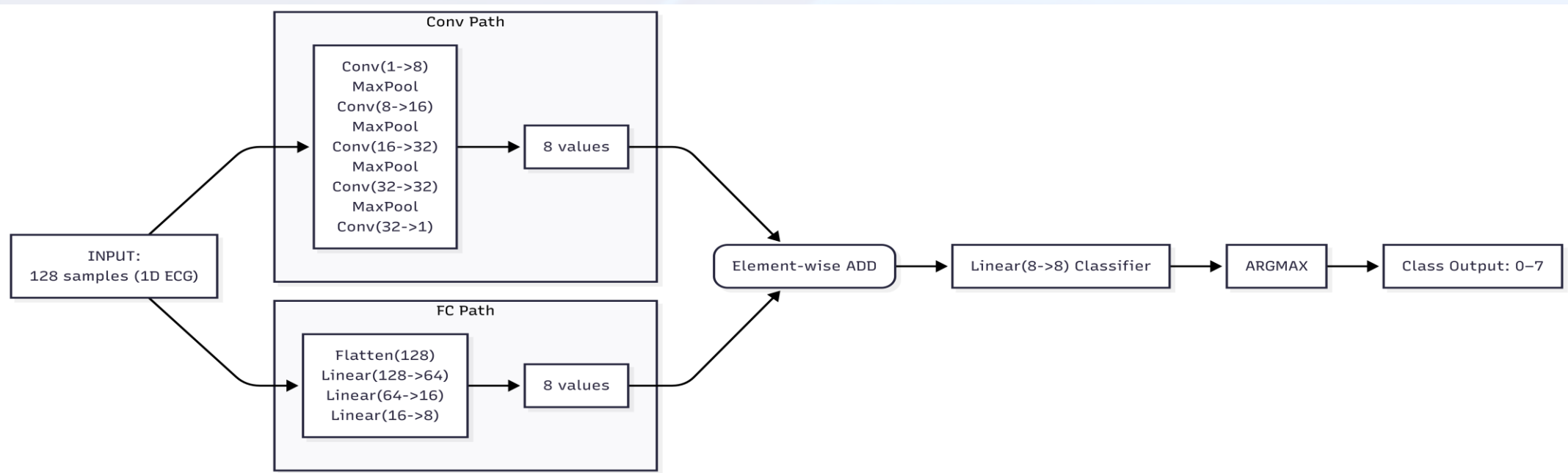


Figure 4: ZolotyNet dual-path architecture

| Parameter      | Value                     |
|----------------|---------------------------|
| Input          | 128 samples @ 360 Hz      |
| Parameters     | ~14,700                   |
| Output classes | 8 arrhythmia types        |
| Quantization   | Q8.8 fixed-point (16-bit) |
| Weight storage | 18 on-chip M4K ROM files  |

### Hardware Implementation

- Time-multiplexed MAC engine:** 1 Conv1D engine handles all 5 conv layers sequentially
- 1 Linear engine** handles all 4 FC layers — keeps logic element usage to only 13%
- 18 on-chip M4K weight ROMs** — zero external memory bandwidth required
- 128-sample input buffer with automatic 12-bit → Q8.8 conversion pipeline
- HD44780 LCD + green LEDs display real-time classification result

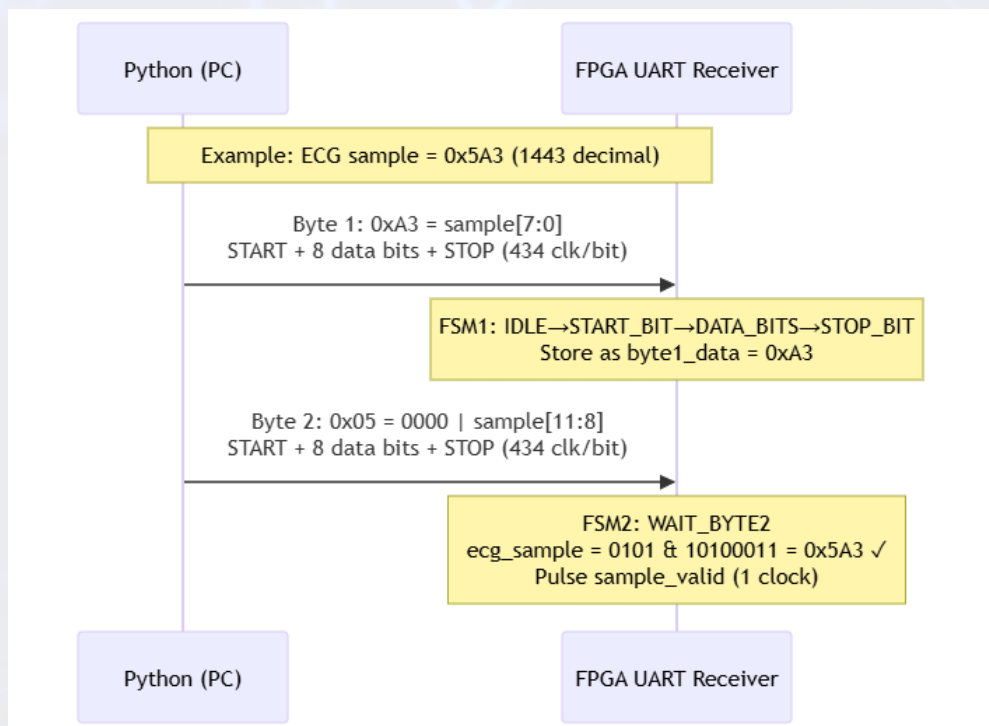


Figure 5: UART two-byte protocol timing diagram

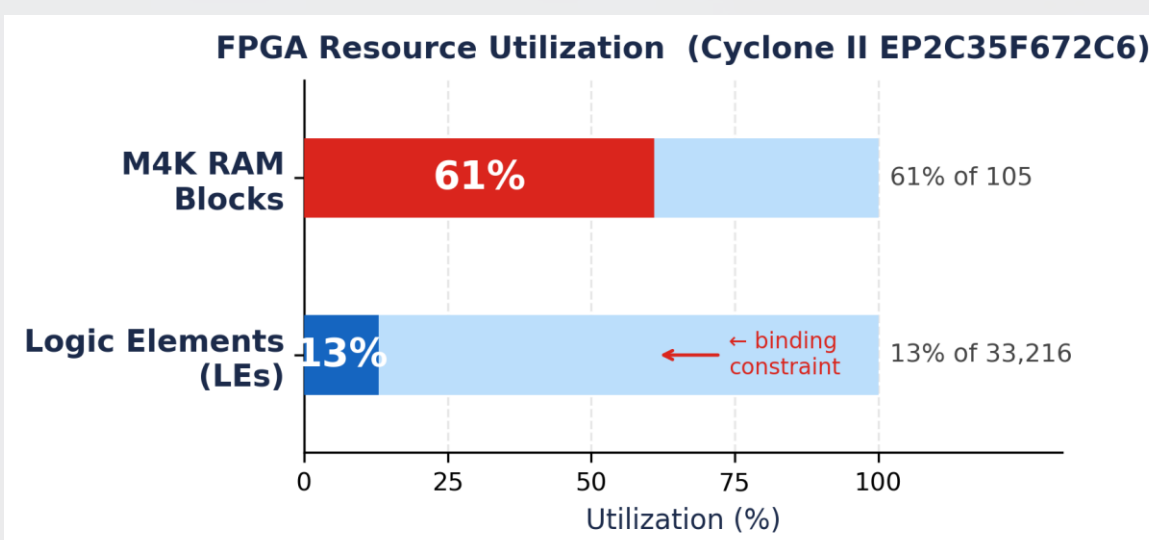


Figure 6: FPGA resource utilization



Figure 7: Board during "Normal" classification

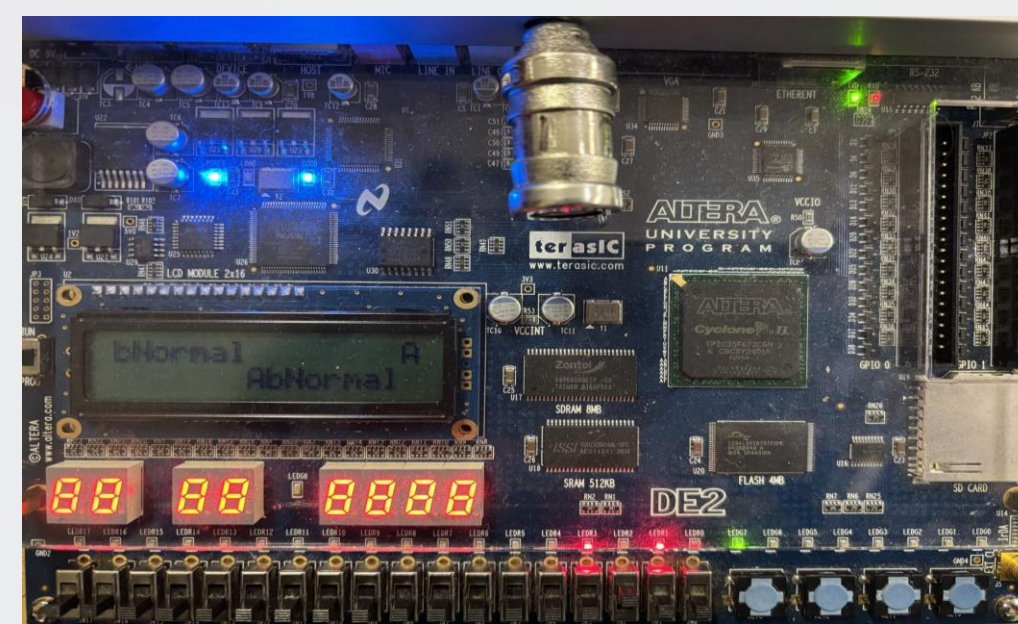


Figure 8: Board during "Abnormal" classification

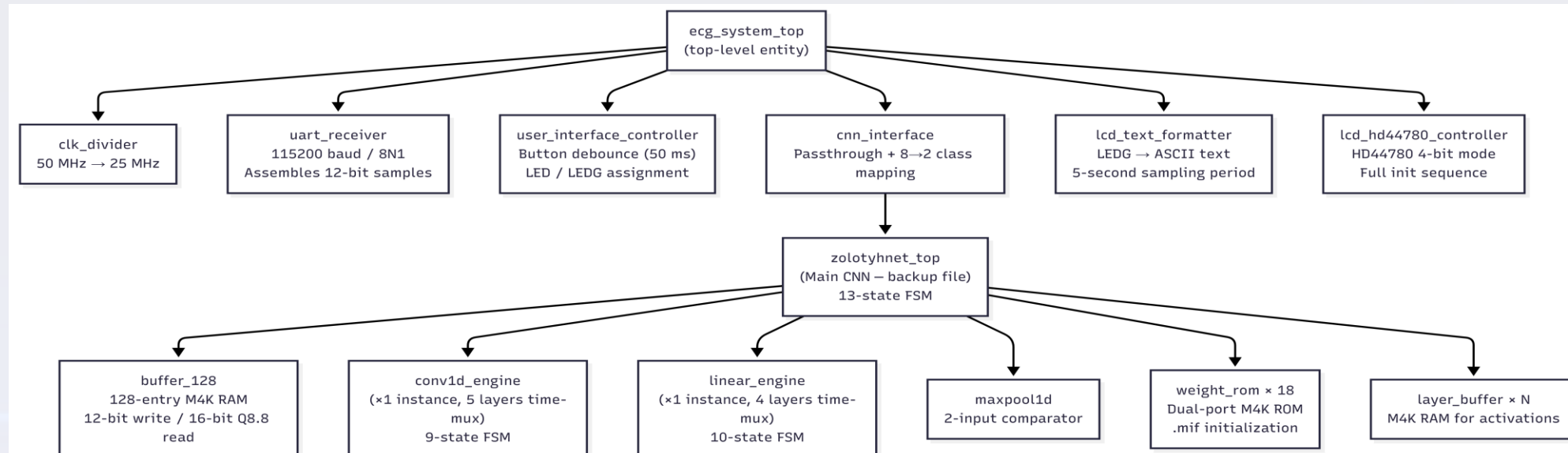


Figure 9: FPGA module hierarchy

## RESULTS

- 95.75% overall accuracy** on 9,368 MIT-BIH validation beats (8,970 correct)
- Normal (N): **99.37%** — near-perfect, dominates dataset
- LBBB (L): **96.41%** | RBBB (R): **94.76%** — well-classified
- Rare classes (E, ! ) underperform due to severe class imbalance (<0.6% of data)
- 2.2 ms** inference per beat window — 160× faster than the 355 ms UART fill time

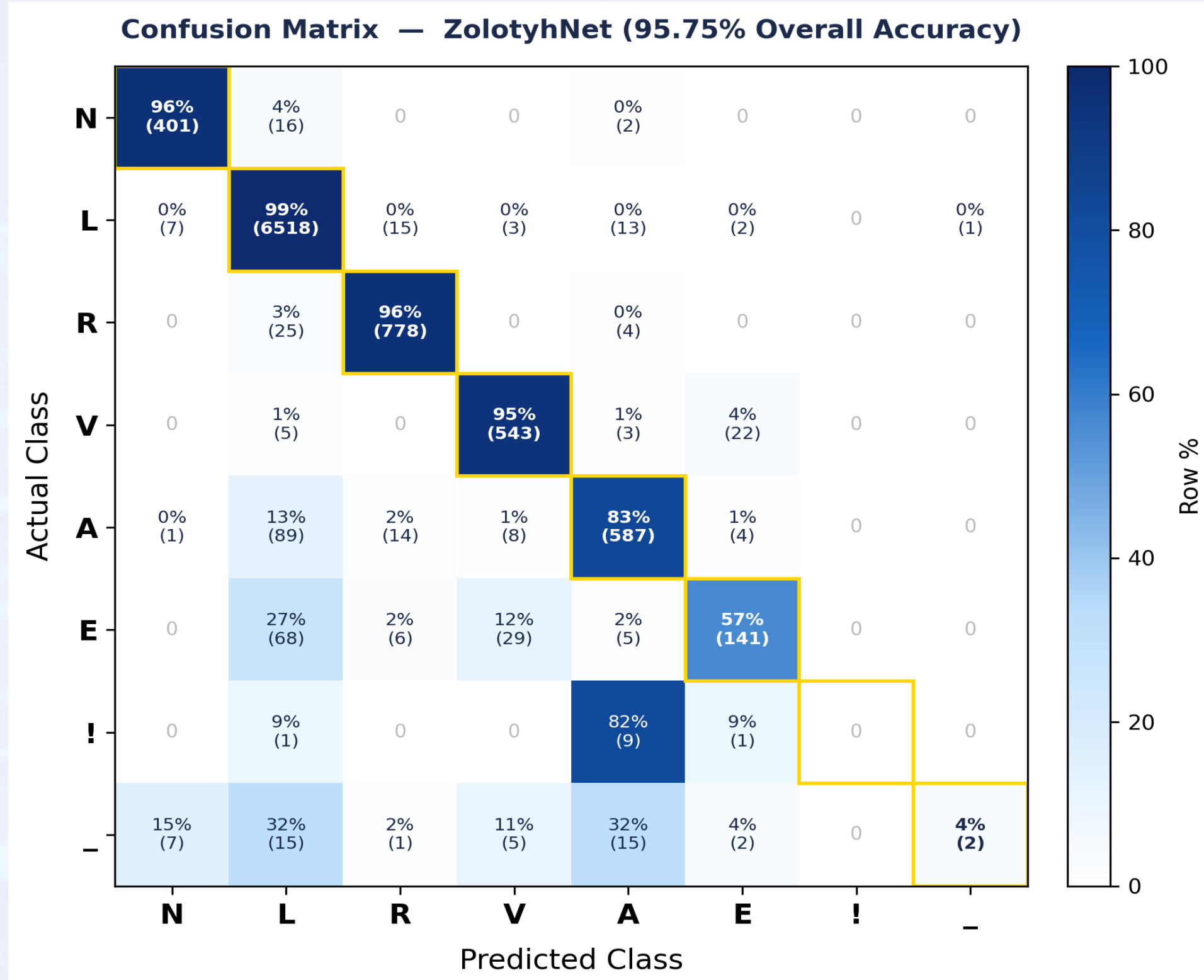


Figure 10: Confusion matrix for ZolotyNet on the 9,368-sample

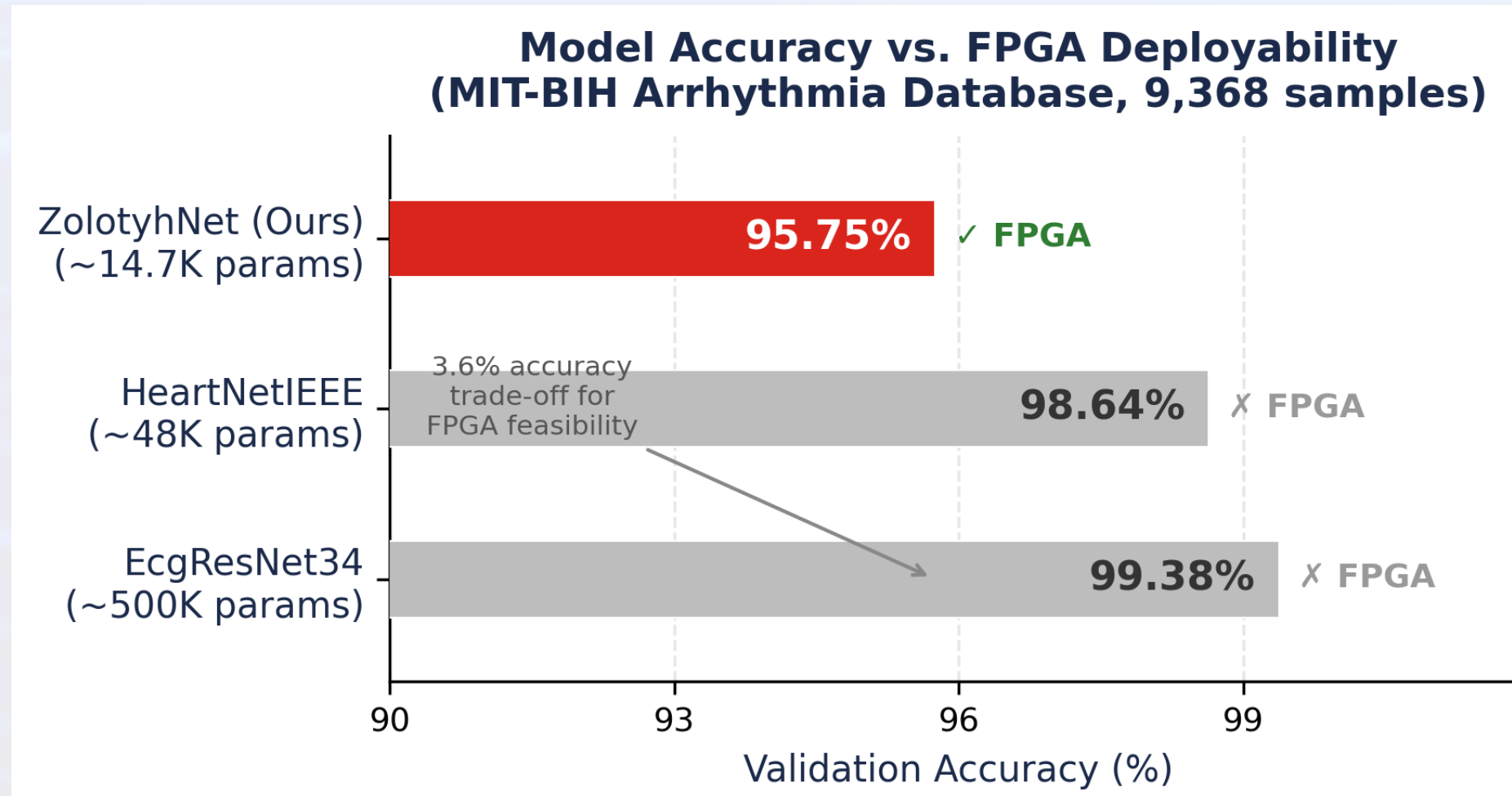


Figure 11: Accuracy comparison.

|                          |                  |                    |
|--------------------------|------------------|--------------------|
| 2.2 ms                   | 360 Hz           | 357 ms             |
| CNN inference per window | UART stream rate | End-to-end latency |

## CONCLUSION

- ✓ Designed ZolotyNet — a 14,700-parameter CNN deployable entirely within on-chip M4K ROM
- ✓ Q8.8 fixed-point quantization pipeline: PyTorch → 18 MIF files → VHDL ROM
- ✓ Time-multiplexed Conv1D + Linear MAC engines: **13% LEs, 61% M4K** on Cyclone II
- ✓ **95.75%** classification accuracy across 8 arrhythmia types
- ✓ **2.2 ms** FPGA inference — 160× within the per-beat timing budget
- ✓ Validated across ~32 different MIT-BIH ECG records: LCD and LED outputs correct